Group Projects

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Introduction

You now will be working on a group project. You can either

- Choose one of the suggestions provided
- Come up with an own idea

If you come up with your own idea, the scope should be similar to those proposed and divided into 3 deliverables. The idea must be accepted by us. For this, please email roland.widmer@students.unibe.ch at least 2 days prior to the deadline of Assignment 7.

General guidelines:

- Make sure that all team members contribute equally.
- Write code in Python scripts that can be run in the terminal. Jupyter notebooks are not allowed.
- Include a `pyproject.toml` (poetry) or `requirements.txt` (pip) file with the package requirements of your project. We recommend the usage of poetry.
- Include a `.gitignore` such that the environment is ignored by Git.
- For every submission, include an updated README file. The README file should contain:
    - How to install
    - How to run
    - Name of all group members
- A naïve user must be able to execute your scripts without reading the code.
- If relevant, database/table(s) necessary to test the project must be included in the submission. Make sure that the data is not too heavy, otherwise use Git LFS. Note: GitHub LFS has usage limits in the free plan (1 GB storage, 1 GB bandwidth/month).
- Without a LICENSE file, the licensing terms are not clearly defined. It is therefore recommended to add a LICENSE file (as an example, you may look at the one from the course repository). Reference to the LICENSE in the README is optional.

1) Air quality dashboard

Create a reporting tool for the air quality that outputs air quality parameters for different countries and cities across time. The data is from the WHO's Global Health Observatory and can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data selection

Allow the user to input one or more countries and a time span and get information about the maximum values of pm10, pm25 and NO2.

Deliverable 3 - Data visualization

Provide visualizations about the trends of air pollution across cities, countries and measures.

2) SBB platform edges

Create a reporting tool for the platform edges around the SBB railway network. The data is from SBB and can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.

3) SBB trains per route per month

Create a reporting tool for the average number of trains per day for each section of the route network and month. The data is from SBB and can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame. Or make it possible such that the user can impute/change the data frame meaningfully.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.

4) Airbnb dataset

Create a reporting tool for the Airbnb accommodations. The data is from Kaggle and can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as data frame. Allow the user to explore information about the dataset and get summary statistics, e.g. on costs, availability, and ratings.

Deliverable 2 - Data selection

Allow the user to input a neighborhood and the room type and get information about the max, min and median costs, with associated information.

Deliverable 3 - Data visualization

Provide visualizations about the cost, reviews and room types, allowing the user to dynamically select the neighborhood.

5) Fitness Tracker

Create a fitness tracker to help users monitor their exercise routines and progress.

Deliverable 1 - Exercise Logging

Allow users to input data about their workouts, including the type of exercise, duration, distance (if applicable), calories burned, … whatever you want. Store this data in a data frame.

Deliverable 2 - Goal Setting and Progress Tracking

Enable users to set personal fitness goals, such as weekly exercise duration, distance, or calories burned. Retrieve and display their progress toward these goals based on their logged workouts.

Deliverable 3 - Data Visualization and Trends

Provide users with visualizations of their exercise data, such as weekly/monthly summaries, progress charts, or comparisons of different exercise types. Allow users to analyze trends in their workout habits over time.

6) Book Manager

Create a book manager to help users store and organize the books they've read and discover new ones.

Deliverable 1 – Books Storage and Organization

Allow users to input and store books with details such as title, author, rating, genre and synopsis. Organize books into different genres, number of pages, topic. You can also add the date they've read the book, reading streaks, etc.

Deliverable 2 – Book Search and Retrieval

Enable users to search for books based on keywords, authors, or genres. Retrieve and display relevant books with their details.

Deliverable 3 - Data Visualization and Analysis

Provide users with visualizations of their book collection, such as a breakdown of books by topic or rating. Allow users to analyze trends in their book preferences, like the rating per author per year. If you like, you can also add additional statistics (e.g., number of books read per month, have they reached their reading goal for the year etc.)

7) Music player

Create a music player to help users store and organize the music pieces they've listen to and discover new ones.

Deliverable 1 – Music database

Allow users to provide paths for the .wav files, with associated information about title, artist, genre, year, etc. Store the database as a dataframe.

Deliverable 2 – Music selection

Enable users to search for music based on artists or genres. Retrieve and display relevant pieces with their details. Users can also choose to perform actions such as amplification or starting playback at a specific point in the song.

Deliverable 3 – Music playing

Users can play a song, and the audio will be played back to them through the speakers. The user can also select to show a plot of the first 10 seconds of the audio file.

8) Podcast Manager

This is a more difficult task aimed at groups wanting to work with external resources/web.
Create a podcast manager to help users organize and listen to their favorite podcasts.

Deliverable 1 - Podcast Subscription and Organization

Allow users to subscribe to podcasts by inputting the podcast's RSS feed URL. Store and organize podcast information, such as title, author, and description. Display a list of subscribed podcasts.

Deliverable 2 - Episode Management

Retrieve and display podcast episodes for each subscribed podcast. Allow users to download, play, pause, and resume episodes, as well as mark episodes as played or unplayed.

Deliverable 3 - Playback Customization and Queue

Enable users to create a playback queue by adding episodes to the queue in their desired order. Allow users to customize playback options such as playback speed and skip intervals for a personalized listening experience.

9) Personal Task Manager

Create a personal task manager to help users organize and track their tasks and deadlines.

Deliverable 1 - Task Input and Organization

Allow users to input tasks with details such as title, description, priority, due date, and status. Organize tasks into categories, like personal, work, or school.

Deliverable 2 - Task Management and Notifications

Enable users to view, edit, and delete tasks. Provide a way to mark tasks as completed, and offer notifications for upcoming deadlines.

Deliverable 3 - Data Visualization and Productivity Analysis

Provide users with visualizations of their task data, such as the number of tasks completed over time, tasks by category, or tasks by priority. Allow users to analyze their productivity trends and identify areas for improvement.

10) Movie Recommendation System

Create a movie recommendation system that helps users discover new movies based on their preferences.

Deliverable 1 - Movie Input and Preferences

Allow users to input movies they have watched along with their ratings. Users can also input their favorite genres, actors, or directors to customize their recommendations.

Deliverable 2 - Movie Recommendation Algorithm

Develop a basic recommendation algorithm that suggests new movies based on user preferences and ratings. Retrieve and display recommended movies along with their details.

Deliverable 3 - Data Visualization and User Trend Analysis

Provide users with visualizations of their movie data, such as a breakdown of movies by genre, actor, or director. Allow users to analyze trends in their movie preferences and discover new content that aligns with their interests.

11) Travel Planner

Create a travel planner to help users organize and plan their trips

Deliverable 1 - Trip Details and Itinerary

Allow users to input trip details, such as destination, dates, transportation, and accommodations. Users can also add activities or points of interest to their itinerary.

Deliverable 2 - Budget Management

Enable users to input their trip budget and track expenses, such as accommodation costs, transportation, meals, and activities. Provide a way to view the remaining budget and recommend adjustments if necessary.

Deliverable 3 - Data Visualization and Trip Analysis

Provide users with visualizations of their trip data, such as a timeline of planned activities or a breakdown of expenses by category. Allow users to analyze their travel patterns and budget usage to optimize future trips.

12) Language Learning Assistant

Create a language learning assistant to help users practice and track their progress in learning a new language.

Deliverable 1 - Vocabulary Management

Allow users to input and store vocabulary words along with their translations, pronunciation, and examples of usage. Organize words into categories, such as topic, difficulty, or part of speech.

Deliverable 2 - Vocabulary Quiz and Practice

Generate quizzes or practice exercises for users based on their stored vocabulary words. Provide immediate feedback on user responses and track their progress over time.

Deliverable 3 - Data Visualization and Progress Analysis

Provide users with visualizations of their learning progress, such as the number of words learned, accuracy in quizzes, or progress in specific categories. Allow users to analyze their strengths and weaknesses to inform future study sessions.

13) Event Planner

Create an event planner to help users organize and manage events, such as parties, conferences, or workshops.

Deliverable 1 - Event Details and Attendee Management

Allow users to input event details, such as date, time, location, and description. Enable users to add and manage attendees, including names, contact information, and RSVP status.

Deliverable 2 - Task Assignment and Timeline

Enable users to assign tasks to attendees or themselves, along with deadlines and status updates. Create a timeline view of the event preparation process, including completed and pending tasks.

Deliverable 3 - Data Visualization and Event Analysis

Provide users with visualizations of event data, such as a breakdown of attendees by category (e.g., professional role, dietary preferences) or task completion status. Allow users to analyze the event planning process and identify areas for improvement.

14) Personal Journal

Create a personal journal application to help users record their thoughts, experiences, and memories.

Deliverable 1 - Journal Entry Management

Allow users to create and store journal entries with details such as title, date, content, and tags or categories. Provide options for text formatting and the ability to add images or links.

Deliverable 2 - Search and Retrieval

Enable users to search for journal entries based on keywords, tags, or date ranges. Retrieve and display relevant entries for review or editing.

Deliverable 3 - Data Visualization and Sentiment Analysis

Provide users with visualizations of their journal data, such as the frequency of entries over time, the distribution of entries by tag or category, or word clouds of commonly used words. Implement a basic sentiment analysis to identify trends in users' emotions or experiences over time.

15) Image Gallery and Editor

Create an image gallery and editor that allows users to store, organize, and edit their images.

Deliverable 1 - Image Upload and Organization

Allow users to upload and store images in various formats (e.g., JPEG, PNG). Organize images into albums or categories, such as by date, event, or subject.

Deliverable 2 - Basic Image Editing

Enable users to perform basic image editing tasks, such as cropping, resizing, adjusting brightness/contrast or applying filters. This can be done via input to the terminal. Provide a preview of the edited image before saving the changes.

Deliverable 3 - Image Metadata and Analysis

Allow users to view and edit metadata associated with their images, such as date, location, and camera settings. Provide visualizations and analysis of image metadata, such as the distribution of images by date or location, or histograms of camera settings (e.g., aperture, shutter speed).

16) Fitbit analysis

Create a reporting tool for the Fitbit data of 30 users that outputs health parameters about sleep, steps, calories, etc. The data are from Kaggle and can be downloaded here.

Deliverable 1 - Data preparation and exploration and combination

Combine the different .csv files into a single data frame, such that all the relevant information can be found in it. Make sure the data can be accessed at different temporal resolutions (minute, hour, or day).

Deliverable 2 - Data selection

Allow users to access the details of the data frame and get summary statistics about health measures, by choosing subjects and time periods.

Deliverable 3 - Data visualization

Provide users with visualizations about health measures across time for different subjects, comparisons between subjects and time periods. Allow users to select the resolution of the data (minute, hour, or day).

17) New York City Payroll data

Create a reporting tool for payroll data of municipal employees of New York City. The data can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame. Or make it possible such that the user can impute/change the data frame meaningfully.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.

18) World happiness report

The World Happiness Report is a publication that presents analyses and rankings of countries based on individuals' self-reported life satisfaction and examines how these ratings correlate with different factors affecting quality of life. The data can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame. Or make it possible that the user can impute/change the data frame meaningfully.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.

19) Net electricity generation in Switzerland

This data set contains data on the total electricity produced domestically, broken down by energy source and time. The data can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame. Or make it possible that the user can impute/change the data frame meaningfully.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.

20) Food facts

Explore nutrition facts from foods around the world with a dataset from Open Food Facts, the World's largest open database of food products. The data can be downloaded here.

Deliverable 1 - Data preparation and exploration

Import the data as a data frame, making sure that columns have the right format. Allow the user to explore information about the dataset.

Deliverable 2 - Data modification and imputation

Allow the user to input new information in the data frame. Or make it possible that the user can impute/change the data frame meaningfully.

Deliverable 3 - Data visualization

Provide visualizations about the dataset, allowing the user to dynamically select the information.